LineaLinkR

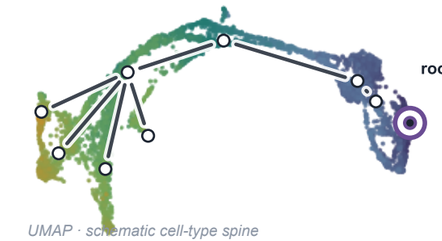
Trajectory-resolved temporal ligand–receptor communication with bifurcation-aware roles

single-cell snapshot → signaling along reconstructed trajectories

01 · INPUT — 1

Snapshot state space

counts → PCA / kNN graph



counts + LR prior

02 · TRAJECTORY — 2

Lineages & time

geodesic pseudotime → bins



- expression-only endpoint-selected attractor root
- Dijkstra geodesic pseudotime
- MST lineages + soft weights → ψ_{il}

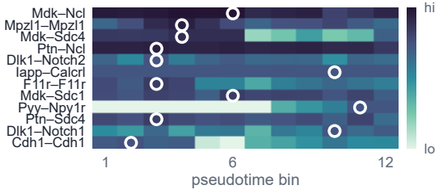
5 lineages recovered

03 · SCORING — 3

Temporal LR programs

peak-lineage envelope · display gaps filled

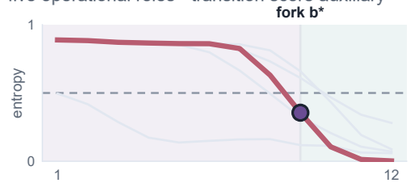
$$A_{LR} = \frac{\sqrt{L \cdot R}}{S + \sqrt{L \cdot R}}$$



04 · INTERPRET — 4

Bifurcation → roles

five operational roles · transition score auxiliary



pre-fork

- Instructive
- Lateral

post-fork

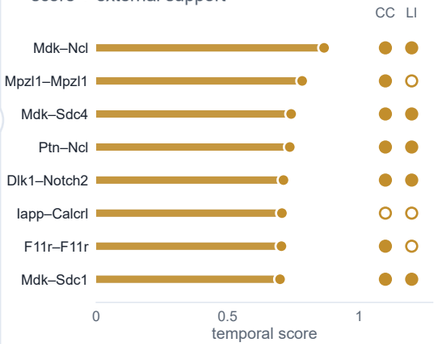
- Reinforcing
- Feedback

- Bystander

05 · EVIDENCE — 5

Ranked programs

score + external support



EVIDENCE

real and simulated data · blind ordering · external biological checks

37	broad evaluation	**registered dataset slots**
24	common-data accuracy	**ordering sets**
4	terminal F1 + ARI	**branch-recovery sets**
11	method-specific adapters	**eligible TI comparators**
5 · 12	timing taxonomy	**operational roles · bins**